

Generation Interconnection System Impact Study Report

For Queue Project Q18-1028

Gardner 230 kV

May 02, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-1028. The Interconnection Customer proposes a 175.2 MW solar photovoltaic facility interconnecting at the Gardner 230 kV substation in the Illinois area. The request was received on July 17, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 313 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 6 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$1,030,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
WAYNE TRACE / DARLINGTON RD 100 kV	DEKALB / RICHLAND 500 kV	107.1	556	5.9
MASON ST 500 kV/230 kV Transformer	BEAVER / TREGO 500 kV	101.5	2,151	11.8
POST ROCK / NEWPORT TIE 230 kV Line	TREGO / BEAVER 500 kV	100.7	2,343	13.1
ADAMS / NEW DEPARTURE 230 kV	TREGO / BEAVER 500 kV	100.5	1,433	7.7
MYRTLE POINT / POST ROCK 230 kV	TREGO / BEAVER 500 kV	100.1	2,391	12.6
MINNEHAHA / SIDNEY A MURRAY JR HYDROELECTRIC SUBSTATION 230 kV	RICHLAND / DEKALB 500 kV	100.1	1,403	16.3

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
WAYNE TRCE / DARLINGTON RD 100 kV	Reconductor line	\$838,000	2% (\$13,000)
MASON ST 500 kV/230 kV Transformer	Replace transformer with higher-rated unit	\$26,639,000	2% (\$415,000)
POST ROCK / NEWPORT TIE 230 kV	Reconductor line	\$6,977,000	2% (\$109,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
ADAMS / NEW DEPARTURE 230 kV	Reconductor line	\$10,962,000	2% (\$171,000)
MYRTLE POINT / POST ROCK 230 kV	Reconductor line	\$10,094,000	2% (\$157,000)
MINNEHAHA / SIDNEY A MURRAY JR YDROELECTRIC 230 kV	Reconductor line	\$10,524,000	2% (\$165,000)
	Total allocated to project		\$1,030,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.